Same Study UID, Soft Delete & Patient ID Conflicts

care_radiology • dcm4chee-arc 5.34.1 • 30 September 2026 • Planning note, no code changes

1. Summary

- **Soft delete** in dcm4chee means **rejecting** the study (IOCM rejection note). Files stay on disk, in the same AE title. Nothing is moved or deleted.
- Soft-deleted studies **stay retrievable** through the existing **IOCM_QUALITY** view. This view is a filter over the same storage, not a move.
- **dcmAcceptConflictingPatientID = YES** lets images of a *different* patient be stored into an existing study. **We recommend not enabling it**. Those images would be filed under the first patient's study.
- The PACS keeps **one study per Study Instance UID**, owned by **one patient**. So "several studies with the same UID, one correct and the rest soft-deleted" only works at the **instance** level, inside one study, for one patient.

2. Current PACS configuration (read from LDAP on 30 Sep 2026)

Setting	Value	Meaning
Rejection note 113001 “Quality” – dcmAcceptPreviousRejectedInstance	**IGNORE**	Re-sending an instance that was already rejected returns success, but the PACS discards it and the instance stays rejected.
Rejection 113001 – dcmDeleteRejectedInstanceDelay	Not set	Quality-rejected instances are **never auto-deleted**. Only “Retention Expired” (113039) has a 1-day delete delay.
AE DCM4CHEE (the one CARE uses)	View **hideRejected**	Rejected instances are hidden from QIDO/WADO.
AE IOCM_QUALITY	View **rejectedForQualityReasons**	Shows **only** quality-rejected instances. The same files, read-only lens.
Rejection note “Revoke Rejection”	REVOKE_REJECTION^99DCM4CHEE	Restores rejected instances.
CARE: CARE_RADIOLOGY_PACS_DELETE_REJECTION_CODE	113001^DCM	CARE already rejects with this code before its (optional) hard delete.

3. What dcmAcceptConflictingPatientID does

This setting applies when an incoming instance belongs to a Study UID the PACS already has, but carries a **different Patient ID** from that study's patient. It is set at device level and can be overridden per AE.

Value	Behaviour	Risk
NO	Refuses the instance with a patient-ID mismatch error. Nothing is stored, and the caller is told why.	**Safe**
YES	Accepts the instance and files it into the **existing study**, under the **existing patient (A)**. It does not create a second study for patient B.	**Patient-safety risk**: B's images appear in A's record. The mismatch is hidden, not resolved.

Key point: the setting does **not** let two patients each have their own copy of a study with the same UID. The Study UID is unique in the archive. YES only turns off the safety check.

4. Scenarios: study S is soft-deleted (rejected 113001) and belonged to Patient A

“Same files” means the same SOP Instance UIDs. “New files” means new SOP Instance UIDs that reuse Study UID S.

#	Upload	With NO	With YES
1	Same files → Patient A	PACS returns success but **discards them** (IGNORE). S stays soft-deleted. CARE sees a "successful" upload with nothing new stored.	Same as NO. The patient check is not involved.
2	Same files → Patient B	Nothing is stored for B. Either the upload is discarded (IGNORE) or refused (patient-ID mismatch). *Which of the two happens needs a test.*	Expected to be discarded (IGNORE). *Needs a test.*
3	New files, same Study UID → Patient A	Stored in study S. S becomes **partly visible**: the new instances show, the old rejected ones stay hidden.	Same as NO.
4	New files, same Study UID → Patient B	**Refused** (patient-ID conflict). Nothing is stored.	**Accepted into A's study S**. B's images are shown as Patient A's.

A rejected study still exists in the PACS database, so it still "owns" its Study UID and Patient A. Soft delete does **not** free the UID for another patient.

5. Getting a soft-deleted study back

- **Restore (recommended):** apply the **REVOKE_REJECTION^99DCM4CHEE** rejection note to study S. Send it through the **IOCM_QUALITY** AE, which is the view that can see the rejected instances. The instances become visible again in DCM4CHEE.
- **Re-upload does not restore**: 113001 is set to IGNORE. Changing it to **RESTORE** would make a re-upload un-reject the instances, but this changes behaviour for every quality rejection.
- **Only viewing, no restore:** query/retrieve via the IOCM_QUALITY AE (QIDO/WADO). OHIF would need a data source that points at this AE.
- **Caveat:** if new files were uploaded into S after the rejection (scenario 3), the restore **merges old and new instances into one study**. Decide which set is "correct" before restoring, or restore selected series/instances only.
- **CARE side:** the matching DicomStudy record (archived/hidden in CARE) must be un-archived as well. Otherwise the PACS and CARE disagree.

6. Recommendation

- Set **dcmAcceptConflictingPatientID = NO**. **Do not set YES.**
- Use rejection 113001 as the soft delete. Do not use the hard-delete step (CARE_RADIOLOGY_DELETE_STUDY_FROM_PACS_ON_ARCHIVE stays False).
- In CARE, before uploading: if the Study UID already exists in the PACS, **including rejected** studies (checked via IOCM_QUALITY), under a **different patient**, block the upload with a clear error. Do not rely on the PACS error.
- Also in CARE: when SOP UIDs match a rejected instance, report the upload as "previously soft-deleted". Do not report 201 Created, because the PACS silently discards it.
- Add a "Restore" action in CARE that sends REVOKE_REJECTION and un-archives the DicomStudy.
- For genuinely wrong-patient studies, correct them with dcm4chee's patient/study correction tools, not by re-uploading.

7. To verify before rollout (test PACS only)

- Scenario 2: whether the result is a silent discard or a patient-ID refusal, and the exact HTTP/DICOM status returned to CARE.
- Scenario 4 with YES, on a test instance only: confirm which patient demographics are shown on retrieval.
- The restore endpoint and AE combination, and restore at series/instance level.
- The status codes CARE receives, so the new error messages map to them correctly.

Configuration values in section 2 were read from the running PACS. The rows marked “needs a test” are expected behaviour from dcm4chee's documented settings and have not yet been confirmed on our instance.